DL1

Membrane Air Separation

Group 4
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Introduction

- The main purpose of this lab was to study the separation process of air (mainly oxygen and nitrogen)
- Separation method used was membrane separation
- Performed separation with both parallel and series flow through the modules
- This lab also served to study the accuracy of theoretical and empirical formulas for calculating the transport coefficients of both O_2 and N_2

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Theory

DL2

- Membrane separation is a function of diffusivity and solubility
- Fick's law describes diffusivity:

$$J_A = D_A\left(\frac{C_{A1} - C_{A2}}{z}\right)$$

- Henry's law describes solubility:

$$C_A = P_A S_A$$

- Terms to describe the separation:
  - Separation efficiency

$$\alpha_{AB} = \frac{Q_A}{Q_B} \qquad \alpha^*_{AB} = \left(\text{[illegible]}\right) \qquad \alpha^{**}_{AB} = \left(\text{[illegible]}\right)$$

  - Recovery

$$A\ Recovery = \left(\frac{V_p C_{Ap}}{V_f C_{Af}}\right)$$

  - Stage cut

$$STAGE\ CUT = \left(\frac{\theta_p}{\text{[illegible]}}\right)$$

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Apparatus

- Main Equipment
  1. Prism separator modules
  2. Pressure regulator
  3. Strain-gage transducer
  4. Flow meter
  5. Filter
  6. Digital pressure readout
  7. O_2 sensors
  8. Air tank
  9. Pressure gauge
  10. 3-way valves

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Slide 1

DL1 98/100
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Slide 3

DL2 Type the equations. Some error happens on this page.
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DL4 -1
Donglin Li, 11/22/2019

DL5 Missing overall transport coefficients.
Donglin Li, 11/22/2019

DL6 -1
Donglin Li, 11/22/2019

Material & Supplies

- Material list
  - Dry compressed air
  - Permeable polymeric film
  - O_2 analyzers
  - Batteries

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Procedure

- Prelab Steps
  1. Familiarize team with apparatus
- Lab Day Steps
  1. Calibrate O_2 sensors
  2. Turn on equipment

  Part A
  1. Set 3-way valves to run air through modules in parallel and test various inlet pressures and flow rates through the modules and record oxygen on both the tube side and the shell side

  Part B
  1. Set 3-way valves to run air through modules in series and test various inlet pressures and flow rates through the modules and record oxygen on both the tube side and the shell side

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Experimental Challenges

- Difficulties
  - Remembering which sensor was for the shell and which sensor was for the tube
- Things to watch out for during the experiment
  - 3-way vales are in the correct orientation
  - The pressure regulator on air tank is set to 140 psig
  - The O_2 sensors are calibrated
- Lessons Learned
  - The experiment is fairly simple to complete if the team understands what needs to be recorded prior to starting the lab
  - Give enough time for O_2 sensors to equilibrate

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Safety

- Caution when adjusting the pressure regulator not to over pressurize system
- Careful not to knock air tank over
- Wear safety glasses at all times

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